

“Two wrongs don’t make a right... unless you are a dragon who uses magical glass.”

Evirir the dragon carried N magical circular plates made out of thin glass, which he will use to build a magical contraption. At first, the plates are black in colour. The plates have an interesting property:

- If two black regions overlap, the overlapping region becomes colourless.
- If a colourless region overlaps with a black region, the region becomes black again.

Precisely, a region becomes colourless if an even number of plates are stacked on it; otherwise, it will remain black.

Unfortunately, having scaly paws, Evirir dropped the plates on the ground and one of the plates is broken! Now he needs to rebuy the broken plate, but he forgot its radius. Luckily, he remembers that when all plates are stacked together, aligned by their center and ordered by their radius (largest radius at the bottom and smallest radius on top), the total area of the black regions is $A\pi$. As long the black area is correct **and all plates have integer radius**, the contraption will work.

Given the radii of the $N - 1$ remaining plates and the value of A , output the radius of the broken plate. If there are multiple possible solutions, output any. It is guaranteed that there is at least one valid integer solution.

Notes:

- The radii of all plates, including the broken plate, must be integers.
- A plate’s radius must be positive.
- The area of a circle with radius r is πr^2 .

Input Format

The input consists of two integers, N and A , and an array R of $N - 1$ integers, where R_i is the radius of the i -th remaining plate.

It is guaranteed that there is at least one valid solution. Also, note that the total black area is $A\pi$, not A .

Constraints:

1. For case 8:

- $N = 10^5$.
- $1 \leq A \leq 10^{10}$.
- $1 \leq R_i \leq 10^5$ ($1 \leq i \leq N - 1$)

2. For all other cases:

- $1 \leq N \leq 5 \cdot 10^4$.
- $1 \leq A \leq 10^{18}$.
- $1 \leq R_i \leq 10^9$ ($1 \leq i \leq N - 1$)

Output Format

Output an integer, the radius of the broken plate. If there are multiple possible solutions, output any.

Sample Input 1

```
N = 4  
A = 10  
R = [1, 4, 2]
```

Sample Output 1

```
3
```

Sample Input 2

```
N = 4  
A = 16  
R = [2, 3, 6]
```

Sample Output 2

Explanation

Sample 1: Suppose the missing radius is 3. The radii sorted in order will be $\{1, 2, 3, 4\}$, so the area of black region is $(4^2 - 3^2 + 2^2 - 1^2)\pi = 10\pi$, which satisfies the value of A . Therefore, a possible value of the missing radius is 3.

Sample 2: Suppose the missing radius is 5. The radii sorted in order will be $\{2, 3, 5, 6\}$, so the area of black region is $(6^2 - 5^2 + 3^2 - 2^2)\pi = 16\pi$, which satisfies the value of A . Therefore, a possible value of the missing radius is 5.

Note that other outputs may also be accepted.

“Two wrongs don’t make a right... unless you are a dragon who uses magical glass.”

Evirir sang naga membawa N piring sakti berbentuk bulat yang dibuat menggunakan kepingan kaca, untuk digunakan oleh beliau bagi membina sebuah peralatan sakti. Pada awalnya, piring tersebut berwarna hitam. Piring tersebut mempunyai sifat yang menarik:

- Jika dua kawasan berwarna hitam bertindih, kawasan yang bertindih tersebut menjadi tiada warna.
- Jika suatu kawasan tiada warna bertindih dengan suatu kawasan hitam, kawasan tersebut menjadi hitam semula.

Secara tepatnya, suatu kawasan menjadi tiada warna jika bilangan piring yang genap bertindan di atasnya; selainnya, kawasan tersebut menjadi hitam.

Malangnya, disebabkan tangannya yang menggerutu, Evirir terhempas piring-piring tersebut ke atas lantai dan satu piring telah pecah! Dia perlu menggantikan piring yang pecah itu, tapi dia terlupa jejarnya. Mujurlah dia teringat apabila semua piring tersebut bertindan, dengan pusat-pusatnya sejajar, dan tersusun mengikut jejari (jejari paling besar di bawah dan jejari paling kecil di atas), jumlah kawasan berwarna hitam ialah $A\pi$. Selagi kawasan berwarna hitam mempunyai luas yang betul **dan semua jejari ialah integer**, peralatan sakti tersebut akan berfungsi.

Diberi $N - 1$ jejari piring yang belum pecah dan nilai bagi A , output jejari bagi piring yang pecah. Jika terdapat beberapa penyelesaian yang mungkin, output salah satu daripadanya. Adalah

dijamin bahawa wujud sekurang-kurangnya satu penyelesaian integer yang sah.

Nota:

- Jejari-jejari semua pinggan (termasuk pinggan yang pecah) mestilah integer.
- Jejari suatu pinggan mesti positif.
- Kawasan suatu bulatan dengan jejari r ialah πr^2 .

Format Input

Input terdiri daripada dua integer, N dan A , dan suatu tatasusunan (array) R yang mengandungi $N - 1$ integer. R_i ialah jejari bagi piring ke- i yang belum pecah.

Adalah dijamin bahawa wujud sekurang-kurangnya satu penyelesaian yang sah. Sila ambil perhatian bahawa jumlah kawasan hitam ialah $A\pi$, bukan A .

Kekangan:

1. Untuk kes 8:

- $N = 10^5$.
- $1 \leq A \leq 10^{10}$.
- $1 \leq R_i \leq 10^5$ ($1 \leq i \leq N - 1$)

2. Untuk kes-kes lain:

- $1 \leq N \leq 5 \cdot 10^4$.
- $1 \leq A \leq 10^{18}$.
- $1 \leq R_i \leq 10^9$ ($1 \leq i \leq N - 1$)

Format Output

Output satu integer, iaitu jejari bagi piring yang pecah tersebut, diasingkan dengan satu ruang. Jika terdapat pelbagai penyelesaian yang mungkin, output salah satu daripadanya.

Contoh Input 1

```
N = 4
A = 10
R = [1, 4, 2]
```

Contoh Output 1

```
3
```

Contoh Input 2

```
N = 4
A = 16
R = [2, 3, 6]
```

Contoh Output 2

```
5
```

Penjelasan

Contoh 1: Andaikan jejari piring yang hilang ialah 3. Jejari-jejari piring tersebut disusun mengikut urutan menaik ialah $\{1, 2, 3, 4\}$. Maka, luas bagi kawasan hitam ialah $(4^2 - 3^2 + 2^2 - 1^2)\pi = 10\pi$ dan ia memenuhi nilai bagi A .

Contoh 2: Andaikan jejari piring yang hilang ialah 5. Jejari-jejari piring tersebut disusun mengikut urutan menaik ialah $\{2, 3, 5, 6\}$. Maka, luas bagi kawasan hitam ialah $(6^2 - 5^2 + 3^2 - 2^2)\pi = 16\pi$ dan ia memenuhi nilai bagi A .

Output lain juga mungkin diterima.

Solution

Note: We will use 0-based indexing here (A_0 is the first number of A).

Sort R in **decreasing** order. Let's put all remaining plates in the way mentioned in the statement. The current total black area is $R_0^2 - R_1^2 + R_2^2 - \dots + (-1)^{N-2} R_{N-2}^2$ (note that R has length $N - 1$). Note: $(-1)^x$ is 1 if x is even and -1 otherwise.

Now, assume the missing plate has radius x . Find the minimum k such that $x \geq R_k$. If such k does not exist (k.e. x is the smallest radius), let $k = N - 1$. We add x at position k of R (such that x is the k -th value). Then, the total black area becomes

$$R_0^2 - R_1^2 + \dots + (-1)^{k-1} R_{k-1}^2 + (-1)^k x^2 + (-1)^{k+1} R_k^2 + \dots + (-1)^{N-1} R_{N-2}^2$$

We want this value to be A . Note that the total black area only depends on the value of k and x . Therefore, we can iterate through all possible k 's and solve for x .

Define

$$\begin{aligned} pre(k) &= R_0^2 - R_1^2 + \dots + (-1)^{k-1} R_{k-1}^2 \\ suf(k) &= (-1)^{k+1} R_k^2 + \dots + (-1)^{N-1} R_{N-2}^2 \end{aligned}$$

Then we want

$$\begin{aligned} pre(k) + (-1)^k x^2 + suf(k) &= A \\ \iff x^2 &= \frac{A - pre(k) - suf(k)}{(-1)^k} \\ \iff x &= \sqrt{\frac{A - pre(k) - suf(k)}{(-1)^k}} \end{aligned}$$

If x is not positive, not an integer, or is not between R_{k-1} and R_k , then this solution is invalid. Otherwise, we can simply output this x as an answer. When implemented naively, however, this method takes $O(N^2)$ time, which is (supposedly¹) too slow for the last cases.

Observe that $pre(k)$ and $suf(k)$ can be precomputed with prefix sums. Define

$$P(i) = R_0^2 - R_1^2 + \dots + (-1)^i R_i^2$$

for $0 \leq i \leq N - 2$ and $P(-1) = 0$. Now we have $pre(i) = P(i)$. If i is odd, then $suf(i) = P(N - 2) - P(k - 1)$, otherwise $suf(i) = -(P(N - 2) - P(k - 1))$. Now we can compute correct x for a particular k in constant time.

Time complexity is $O(N)$.

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1. It is later found out that the tests are too weak, so brute force solutions (e.g. did not use prefix sums) that skip solving for x when the answer is invalid passed. [↩](#)